

Here is the step-by-step process to install Quantum ESPRESSO.

- 1) `sudo apt-get update`
- 2) `sudo apt-get upgrade`
- 3) `sudo apt-get install git`
- 4) `sudo apt install --no-install-recommends \`
 `autoconf \`
 `build-essential \`
 `ca-certificates \`
 `gfortran \`
 `libblas3 \`
 `libc6 \`
 `libfftw3-dev \`
 `libgcc-s1 \`
 `liblapack-dev \`
 `wget`
- 5) `apt search libelpa` (To check which version of libelpa is available in your ubuntu)
- 6) `sudo apt install --no-install-recommends \`
 `libopenmpi-dev \`
 `libscalapack-openmpi-dev \`
 `libelpa19` (libelpa19 for ubuntu 24.04)
- 7) `wget https://gitlab.com/QEF/q-e/-/archive/qe-7.3/q-e-qe-7.3.tar.gz`
- 8) `tar -xvzf q-e-qe-7.3.tar.gz`
- 9) `cd q-e-qe-7.3/`
- 10) `./configure` (For parallel compilation: `./configure MPIF90=mpif90 F90=gfortran CC=mpice`)
- 11) `make all`
- 12) `cd bin`
- 13) `pwd` (copy `/home/manjil/q-e-qe-7.3/bin`)
- 14) `nano ~/.bashrc`
- 15) scroll to the end of the file and add the following
 `export PATH=$PATH:/home/manjil/q-e-qe-7.3/bin`
 Press ctrl + X to exit
 Press Y to confirm changes
 Press Enter to save
- 16) `source ~/.bashrc` (Reload the bashrc to apply the changes)
 Note: You could also use:
 `echo 'export PATH=$PATH:/home/manjil/q-e-qe-7.3/bin' >> ~/.bashrc`